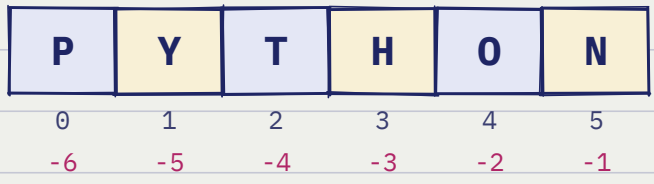
STRINGS: TEXT YOU CAN SLICE

A `str` is text. Strings are **immutable**: methods return a new string.

1) Index and slice



Example :

```
word = "PYTHON"
print(word[0], word[-1])
print(word[0:3])
print(word[2:])
print(word[::-1])
```

Output :

```
P N
PYT
THON
NOHTYP
```

a slice excludes the stop index; a step of -1 reverses; lists use the same rules

2) Useful string methods

Method	Example	Result
.lower() .upper()	"Python".upper()	'PYTHON'
.title()	"rahul sharma".title()	'Rahul Sharma'
.strip()	" hi ".strip()	'hi'
.replace(old, new)	"cat".replace("c", "b")	'bat'
.split()	"red blue".split()	['red', 'blue']
", ".join(list)	", ".join(["red", "blue"])	'red, blue'
.find(text)	"Python".find("th")	2
.endswith()	"a@b.com".endswith(".com")	True

3) Check and format

Example :

```
first_name = "Priya"
last_name = "Patil"
marks = 92.678
full_name = first_name + " " + last_name
print(f"Student: {full_name}")
print(f"Marks: {marks:.2f}")
print("Python123".isalnum(), "12345".isdigit())
```

Output :

```
Student: Priya Patil
Marks: 92.68
True True
```

LISTS: THE MOST USED COLLECTION

A `list` holds many values in `[ ]`. It is ordered, **mutable** and allows duplicates.

1) Create, index and update

Example :

```
students = ["Rahul", "Priya", "Amit", "Priya"]
print(students[0], students[-1])
print(students[:-2])
print(students[::-1], len(students))
marks = [78, 92, 85]
marks[0] = 80
print(marks)
```

Output :

```
Rahul Priya
['Rahul', 'Priya']
['Priya', 'Amit', 'Priya', 'Rahul'] 4
[80, 92, 85]
```

2) Add and remove items

Method	What it does	Result
`.append("Amit")`	adds one item at the end	`['Rahul', 'Priya', 'Amit']`
`.insert(1, "Ram")`	adds at index 1	`['Rahul', 'Ram', 'Priya']`
`.extend(["Isha"])`	adds each item of another list	`['Rahul', 'Priya', 'Isha']`
`.remove("book")`	removes the first match only	`['pen', 'pencil', 'book']`
`.pop()`	removes and returns the last item	`'book'`
`del items[0]`	deletes by index	`['book', 'pencil', 'book']`

3) Sort, copy and built-in functions

Example :

```
marks = [85, 72, 96, 81]
print(sorted(marks))
print(marks)
marks.sort()
backup = marks.copy()
backup.append(100)
print(marks, backup)
print(len(marks), min(marks), max(marks), sum(marks))
```

Output :

```
[72, 81, 85, 96]
[85, 72, 96, 81]
[72, 81, 85, 96] [72, 81, 85, 96, 100]
4 72 96 334
```

sort() changes the list and returns None; sorted() gives a new list

Sets & Choosing A Type

A `set` stores **unique** items: no indexing and no guaranteed order.

1) Create a set

Example :

```
subjects = {"Python", "Java", "Excel", "Python"}
print(sorted(subjects))
print("Java" in subjects, len(subjects))
print(type({}), type(set()))
```

Output :

```
['Excel', 'Java', 'Python']
True 3
<class 'dict'> <class 'set'>
```

☆ WATCH OUT

- `{}` makes an empty **dict**, not a set. Use `set()` for an empty set.

2) Add, remove and set operations

Method	What it does
`add(item) / update(items)`	add one item / many items
`discard(item)`	removes it; no error if missing
`remove(item)`	removes it; KeyError if missing

Example :

```
A = {1, 2, 3, 6}
B = {3, 4, 5, 6}
print(A | B)
print(A & B)
print(A - B, B - A)
names = ["Priya", "Amit", "Priya", "Rahul"]
print(sorted(set(names)))
```

Output :

```
{1, 2, 3, 4, 5, 6}
{3, 6}
{1, 2} {4, 5}
['Amit', 'Priya', 'Rahul']
```

union means either, intersection means both, difference means first only

3) Choosing the right type

Type	Ordered?	Mutable?	Duplicates?	Best for
`str`	Yes	No	Yes	text
`list`	Yes	Yes	Yes	a changeable sequence
`set`	No	Yes	No	unique values, set operations

Mistakes & Practice

Find the bug in each line, then try the tasks before comparing with the solutions.

1) Spot the bug

Buggy line	Problem	Fix
`word[0] = "J"`	strings are immutable	`word = "J" + word[1:]`
`marks.append(4, 5)`	append() takes one item	`marks.extend([4, 5])`
`new = marks.sort()`	new is None	`new = sorted(marks)`
`skills.remove("Go")`	KeyError if missing	`skills.discard("Go")`
`empty = {}`	this is a dict	`empty = set()`

2) Practice tasks

Task 1 Clean a name.

Solution :

```
print(input("Name: ").strip().title())
```

```
Name:   asha PATIL
Asha Patil
```

Task 2 Replace and split.

Solution :

```
s = "Computer Science"
print(s.replace("Science", "Programming"))
print("Python,SQL,Excel".split(","))
```

```
Computer Programming
['Python', 'SQL', 'Excel']
```

Task 3 List edits.

Solution :

```
subjects = ["Maths", "Python", "SQL", "Art", "Music"]
subjects.append("Excel")
subjects.insert(1, "Git")
subjects.remove("Art")
print(subjects)
```

```
['Maths', 'Git', 'Python', 'SQL', 'Music', 'Excel']
```

Task 4 Highest, lowest, average.

Solution :

```
marks = [78, 92, 85, 66, 91]
print(max(marks), min(marks), sum(marks))
print(sum(marks) / len(marks))
```

```
92 66 412
82.4
```

Task 5 Both, either, Python only.

Solution :

```
py = {"Rahul", "Priya", "Amit"}
sql = {"Priya", "Neha", "Amit"}
print(sorted(py & sql))
print(sorted(py | sql))
print(sorted(py - sql))
```

```
['Amit', 'Priya']
['Amit', 'Neha', 'Priya', 'Rahul']
['Rahul']
```

Task 6 Unique and discard.

Solution :

```
subjects = {"Maths", "Python", "SQL", "Python"}
subjects.add("Art")
subjects.discard("SQL")
print(sorted(subjects))
```

```
['Art', 'Maths', 'Python']
```

Recap

str for text, **list** for an ordered collection that changes, **set** for unique items and comparisons.